

Linear TIME: exact ridge forecasting

A semantic panel contains one physical quantity observed for N aligned members. For origin i , histories are $X_i \in \mathbb{R}^{N \times L}$ and targets are $Y_i \in \mathbb{R}^{N \times H}$. Distinct physical quantities form separate panels; repeated users or sensors remain rows of the same panel.

Per member, an optional affine transform uses history mean μ_{in} and population standard deviation σ_{in} . Its offset/scale pair is one of $(0, 1)$, $(\mu, 1)$, $(\mu, \sigma + \epsilon)$, or $(0, |\mu| + \epsilon)$. Optional endpoint-padded moving-average decomposition concatenates residual and trend histories.

The three model modes are

Mode	design	output
independent	one L -feature design per member	one H -output head per member
joint	flattened NL design	one NH -output head
shared	pooled member windows with L features	one shared H -output head

The default is joint. User generalization uses shared mode so coefficients fitted on seen members apply to unseen members.

For design row z_i , normalized target t_i , loss scale d_i , output scale s_i , and $q_i = (s_i/d_i)^2$, each output solves

$$\min_b \frac{1}{M} \sum_{i=1}^M q_i (z_i b - t_i)^2 + \alpha \|Pb\|_2^2,$$

where P excludes the intercept. The declared loss scale is 1 for MSE, $\sigma + \epsilon$ for NMSE, and $|\mu| + \epsilon$ for relative MSE. Sufficient statistics give the exact primal system

$$(Z^\top QZ/M + \alpha P^\top P)b = Z^\top Qt/M.$$

For the default raw-MSE joint model, large designs use the equivalent centered dual system

$$(Z_c Z_c^\top / M + \alpha I)a = Y_c, \quad \hat{Y} = \bar{Y} + (Z - \bar{Z})Z_c^\top a / M.$$

This avoids materializing an infeasible $(NL) \times (NH)$ coefficient matrix without approximation. Five candidate alphas are scored on validation; the selected system remains fitted on training labels only.

MAE and MSE average finite horizon labels per window/member. NMSE and relative MSE divide MSE by their squared loss scales. Here **rmse** denotes relative MSE. MASE divides MAE by the mean finite seasonal difference over all original history preceding the origin; missing or zero scale is undefined. Seasonal Naive repeats the last complete-prefix season independently of L . Scaled MASE uses its matched population/task mean. W10 is the mean of the worst $\lceil 0.1U \rceil$ finite member means among U members.

Only user generalization is random. Its seeds 0, 1, 2 define the complete 80/20 member partition. For seed scores m_s , reports give their arithmetic mean and sample standard deviation ($ddof = 1$), displayed as mean $\pm$ SD.

Implementation. Separate data, ridge, pipeline, results, and visualization packages own these stages. Each report consumes the completed tasks named by one exact launch manifest.